

AvanzaMath — Product Requirement Document

Elementary Mathematics Learning Platform Rooted in Hispanic Culture

TARGET AUDIENCE

Grades K–5 & Bilingual Families

PRIMARY STACK

Offline-First PWA
(HTML/CSS/JS)

AUDIO & SPEECH

Native Web Speech API
(ES/EN)

COMPLIANCE

COPPA, FERPA & Title I Ready

1. Executive Summary & Vision

AvanzaMath is an equity-centered, dual-language, gamified elementary math learning platform designed for elementary students in grades K–5 from low-income, predominantly Hispanic neighborhoods.

Instead of abstract, disconnected drills, AvanzaMath anchors arithmetic (**Addition, Subtraction, Multiplication, Division**) in familiar neighborhood cultural traditions: counting fresh fruits at *El Puesto de Frutas*, baking conchas at *La Panadería*, calculating change at *El Mercado*, and dividing treats from *La Piñata*.

🌟 Core Pillars of AvanzaMath:

- **Culturally Resonant Contexts:** Everyday scenarios Hispanic elementary kids immediately recognize.
- **Bilingual Segmented Toggle** [🇲🇽 ES | 🇺🇸 EN]: Instant dual-language switching with native Latin American Spanish and English TTS audio.
- **Zero-Friction Student Onboarding:** Avatar companion selection and 4-word anonymous recovery codes (zero student PII required).
- **Family Math Center:** Spanish-first parent progress tracking, zero-password QR pairing, and WhatsApp shareable progress cards.

2. Problem Statement & Demographic Context

- **Curriculum Disconnection:** Standard word problems rely on unfamiliar cultural contexts rather than real-world community life.
- **Language Friction for Dual-Language Learners:** ELL students often understand math logic but struggle with English-heavy standardized phrasing.
- **Digital & Hardware Constraints:** Low-income families often rely on prepaid data or shared devices with intermittent connectivity.
- **Parental Barriers:** Spanish-speaking parents want to help with math but face complex, English-only login portals.

3. Target User Personas

PERSONA	PROFILE & CONTEXT	KEY NEEDS & AVANZAMATH SOLUTION
Mateo (Student) Age 8, 3rd Grade	Bilingual home; loves soccer (*fútbol*), video games, and bakery visits with his grandfather.	Visual, game-like practice (*Panadería* arrays, *Piñata* candy division) with Tito the Hummingbird companion and non-punitive hints.
Carmen (Caregiver) Working Mother	Primary language is Spanish. Shares one smartphone with Mateo and his sister.	Zero-password QR pairing to view Mateo's progress in Spanish, WhatsApp updates, and dinner-table math retos.

Maestro Morales (Teacher) Dual-Immersion Educator	Teaches 28 elementary students; 80% are Hispanic dual-language learners.	Supplementary classroom math stations, zero login friction, and printable bilingual offline worksheets.
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4. Curriculum & Pedagogical Framework

AvanzaMath implements the **Concrete-Pictorial-Abstract (CPA)** approach combined with **Culturally Responsive Teaching (CRT)**.

LEVEL	GRADE	ARITHMETIC DOMAIN	CORE COMPETENCIES	HISPANIC CULTURAL THEME MODULE
Nivel 1	K – 1st	Number Sense & Add/Sub	Counting 0–20, Ten-Frames, Number Bonds, Single-digit (+ / –)	El Puesto de Frutas: Counting mangos, avocados, papayas, and oranges on ten-frames.
Nivel 2	2nd – 3rd	Multi-digit Add/Sub & Multiplication	Regrouping, Arrays, Skip Counting, Times Tables (1–10)	La Panadería: Baking conchas and empanadas in rows $\$ \times \$ \text{ cols}$; El Mercado: Dollars & coins (\$1, \$5, \$10, \$20, 25¢).
Nivel 3	4th – 5th	Division & Equal Sharing	Long Division, Factors, Multiples, Equivalent Fractions	La Piñata: Fair candy sharing into party favor bags (*bolsitas*); La Cocina: Scaling family recipes.
Todos	K – 5th	Mental Math Review	All 4 Operations (+, –, ×, ÷), Speed, Pattern Recognition	Lotería Matemática: Math Bingo with bean stamps, operation filters (+, –, ×, ÷), and vocal card caller (*El Cantor*).

5. Hispanic Cultural Integration Modules ("El Sistema Nuestra Cultura")

 **Everyday Community & Celebration Quests**

- **La Panadería de Don Carlos:** Multiplication as baking arrays (3 rows of 4 conchas \$= 12\$).
- **El Mercado y Tiendita:** Budgeting purchases and calculating exact change from \$10 or \$20 bills.
- **La Piñata de Cumpleaños:** Dividing piñata candy batches equally into party bags (*bolsitas*).
- **Lotería Matemática:** Mental math Bingo review across addition, subtraction, multiplication, and division.
- **Héroes STEM:** Real-world missions inspired by astronaut Dr. Ellen Ochoa and calculus educator Jaime Escalante.

6. Functional Requirements & Feature Specifications

FR-01: Welcome Landing & Quick Lesson Prep Hub (Página de Bienvenida)

- Dedicated welcome home view featuring *Tito el Colibrí* and direct CTAs to start math quests or play Lotería.
- **Quick Prep Lessons:** Conceptual summaries and tips for Ten-Frames, Arrays, Money Math, and Division.
- **Curated Video Linkages:** Direct links to Khan Academy (English and Spanish) and Math Antics video lessons.

FR-02: Student Onboarding, Identity & Avatar Profile System (Registro sin Fricción)

- **Zero-PII Compliance:** No email, password, or student personal identifying information collected.
- **Avatar Selection:** Students pick cultural companions (*Tito el Colibrí 🦋*, *Sol el Axolote 🐸*, *Luna la Mariposa 🦋*).
- **4-Word Magic Recovery Code & QR Pasaporte:** Anonymous code (e.g., Colibri–Pan–Sol–24) and printable QR passport to resume progress across school Chromebooks and home tablets.

FR-03: Interactive Digital Manipulatives

- **Ten-Frames with Matching Fruit Emojis:** 🥭 mangos, 🥑 avocados, 🍊 oranges, 🥥 papayas for hands-on counting.
- **Panadería Baking Trays:** Multi-flavor concha arrays (pink, chocolate, vanilla) for multiplication grids.
- **Cash Register & Coin Counter:** Interactive dollar bills (\$1, \$5, \$10, \$20) and change calculation.

FR-04: Friendly Hispanic Mascot & "Ganas" Gamification

- *Tito el Colibrí* provides cheerful growth-mindset coaching and cultural role model facts on every screen.
- **Non-Punitive Pedagogy:** Incorrect answers trigger gentle scaffolding hints; no loss of lives or lockouts.
- Practice earns ****Puntos Ganas**** (★) to celebrate effort and consistency.

FR-05: Instant Dual-Language Engine & Audio Synthesis

- Segmented header toggle [🇲🇽 ES | 🇺🇸 EN] with instant bilingual translation and state persistence.
- Native Web Speech API text-to-speech with Latin American Spanish (es-MX, es-US, es-419) and English (en-US).
- Symbol normalizer translates operators naturally (Spanish: + *más*, - *menos*, × *por*, ÷ *entre*; English: + *plus*, - *minus*, × *times*, ÷ *divided by*).
- Dedicated Auto-Read toggle (🔊 Auto: OFF by default) and manual 🗣️ Read Aloud buttons.

FR-06: Offline-First Progressive Web App (PWA)

- 100% of arithmetic practice, audio syntheses, and mini-games function without internet via Service Worker.
- **Network-First Caching Strategy:** Delivers fresh live updates while providing offline fallback.
- Total initial asset payload strictly < **2.5 MB** for instant loading on budget mobile networks.

FR-07: Family Math Center (El Portal Familiar) & Parent-Child Association

- **Shared Home Device View:** Direct access to mastery progress and Ganas Stars on shared family phones/tablets.
- **Zero-Password QR & Magic Link Pairing:** Parents scan the child's QR code or enter their 4-word code (Colibrí-Pan-Sol-24) to link their phone to a read-only progress dashboard.
- **WhatsApp Progress Sharing:** Encodes achievements into a pre-filled celebratory WhatsApp message with voiceover link.
- **Retos en Familia:** Weekly dinner-table or grocery store math prompts in Spanish.

7. Non-Functional Requirements (NFR)

ID	CATEGORY	REQUIREMENT DESCRIPTION	TARGET METRIC
NFR-01	Performance	First Contentful Paint (FCP) on budget mobile phones and Chromebooks.	$\leq 1.2\text{ s}$
NFR-02	Bundle Size	Total compressed asset payload (HTML, CSS, JS, SVGs, audio synthesis).	$\leq 2.5\text{ MB}$
NFR-03	Accessibility	WCAG 2.1 Level AA compliance, large touch targets ($\geq 48\text{ px}$).	100% Lighthouse Score
NFR-04	Device Compatibility	Seamless execution on budget Android devices, iPad iOS, and ChromeOS.	> 99% Compatibility
NFR-05	Privacy & Legal (COPPA/FERPA)	Zero student PII collected. No emails, passwords, ad trackers, or behavioral profiling.	100% Compliance
NFR-06	Offline Capability	All core practice problems and audio TTS must operate with zero active Wi-Fi.	100% Offline Functional

8. UX/UI Design & Aesthetic Guidelines

 **Hispanic Folk-Art Design System:**

- **Color Palette:** Fiesta Terracotta (#D9531E), Marigold Gold (#F59E0B), Avocado Jade (#10B981), Deep Indigo (#1E293B), and Warm Parchment (#FDFBF7).
- **Typography:** *Outfit* for friendly, expressive headings; *Plus Jakarta Sans* / *Inter* for clear numeral glyphs.
- **Chunky Tactile Touch Targets:** Minimum $48\text{px} \times 48\text{px}$ interactive buttons for small hands on touchscreens.

9. Technology Architecture & Implementation Stack

LAYER	TECHNOLOGY	PURPOSE & ARCHITECTURAL JUSTIFICATION
Frontend Shell	Semantic HTML5 & Vanilla Modern CSS	Zero build-step overhead; instant load speed and maximum responsiveness across devices.
Application Logic	ES6+ Modular JavaScript	Lightweight component architecture without heavy framework dependencies.
Audio & Voice	Web Speech API & Web Audio API Synthesizer	100% free, zero-network-latency bilingual TTS and procedurally generated sound FX.
Offline Cache & State	Service Worker & LocalStorage / IndexedDB	Network-First PWA caching strategy with offline mastery tracking.

10. Risk Assessment & Mitigation Matrix

RISK FACTOR	IMPACT	LIKELIHOOD	MITIGATION STRATEGY
Low Household Bandwidth	High	High	Network-first Service Worker caching and keeping initial bundle $\leq 2.5\text{MB}$.
Student Login Friction	High	Medium	Zero-PII onboarding with avatar selection and 4-word anonymous magic codes.
Speech API Accent Variations	Medium	Medium	Prioritize standard Latin American Spanish (es-MX, es-US, es-419) with clear text fallback.
Math Anxiety	High	Low	Non-punitive micro-rewards, cheerful mascot encouragement, and visual step-by-step hints.
Parent Adoption Barriers	Medium	Medium	Zero-password QR pairing, frictionless WhatsApp sharing, and practical dinner-table retos.

11. Verification & Acceptance Criteria

1. **Accessibility:** All buttons, inputs, and questions have explicit `aria-label` attributes and pass 100% WCAG 2.1 AA audits.
2. **Instant Language Switching:** Segmented language switcher toggles between English and Español under 50ms with full persistence across page refreshes.
3. **100% Offline Capability:** Disconnecting Wi-Fi during a math quest allows full continuation of problem solving, animations, audio, and local streak tracking.

4. **COPPA / FERPA Compliance:** Zero collection of student emails, passwords, or personal identifying data.
5. **Cultural Grounding:** Every grade level includes distinct, everyday Hispanic cultural scenarios (*La Panadería, El Mercado, La Piñata, Lotería, or Héroes STEM*).